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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 1

Design and Develop Model

COURSE NAME: Deep Learning
SLOT: D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **Ch. Venkata Lakshmi Hymavathi (192425397)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Hymavathi

Name of the student:

Ch. Venkata Lakshmi Hymavathi

Criteria	Problem Understanding & Requirement Analysis (2)	Model Architecture & Design (2.5)	Application of Deep Learning Techniques (2.5)	Model Development & Implementation Strategy (2)	Performance Analysis & Justification (1)	Total (10)
Weightage	20 %	30%	25%	15 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

1. Design and develop a model for a hospital has only 2,000 labeled X-ray images for classifying images into normal and abnormal categories. A pre-trained ResNet model is available. Design a CNN-based transfer learning model by identifying which layers should be frozen, which layer should be replaced, and how the model should be trained.

Answer:

Problem:

A hospital has only **2,000 labeled X-ray images** for classifying images into **Normal** and **Abnormal** categories. A pre-trained ResNet model is available.

Model Architecture

- Input: X-ray image, e.g., $224 \times 224 \times 3$
- Pre-trained **ResNet50** as the feature extractor.
- Freeze most/all ResNet convolutional layers initially.
- Remove the original ImageNet classification layer.
- Add:
 - Global Average Pooling
 - Dense layer with ReLU
 - Dropout
 - Final Dense layer with **1 neuron + Sigmoid**

Architecture:

X-ray → ResNet50 → Global Average Pooling → Dense → Dropout → Sigmoid → Normal/Abnormal

Training Strategy

1. Resize and normalize X-ray images.
2. Apply data augmentation such as rotation, flipping and zooming where medically appropriate.
3. Freeze the ResNet layers.
4. Train only the newly added classification layers.
5. After initial training, unfreeze the last few ResNet blocks.
6. Fine-tune using a **very small learning rate**.
7. Use binary cross-entropy loss and Adam optimizer.

Performance Evaluation

Use:

- Accuracy
- Precision
- Recall/Sensitivity
- F1-score
- Confusion matrix
- ROC-AUC

Justification:

Transfer learning is suitable because only 2,000 labeled images are available. The pre-trained ResNet already contains useful visual features, reducing training time and overfitting. Fine-tuning the deeper layers allows the model to adapt these features to X-ray images.

2. An agricultural organization wants to identify **five types of plant diseases** from leaf images. Since the dataset contains limited training samples, the organization wants to reuse features efficiently. **Design a DenseNet-based classification model** and explain how dense connections can help the model reuse features during classification.

Answer:

The agricultural organization wants to classify **five plant diseases** from leaf images with limited training samples.

Model Architecture

Use a pre-trained **DenseNet121**.

Architecture:

Leaf Image → DenseNet121 → Global Average Pooling → Dense Layer → Dropout → Softmax → 5 Diseases

Working

DenseNet connects each layer to **all subsequent layers within a dense block**.

For example:

Layer 1 → Layer 2 → Layer 3 → Layer 4

Layer 4 receives feature maps from Layers 1, 2 and 3.

Feature Reuse

Dense connections allow:

- Earlier features to be reused by later layers.
- Better gradient flow.
- Reduced vanishing-gradient problems.
- Fewer redundant features.
- Efficient learning with limited data.

Training Strategy

1. Resize leaf images to 224×224 .
2. Normalize pixel values.
3. Apply augmentation such as rotation, scaling and flipping.
4. Load pre-trained DenseNet121.

5. Freeze initial layers.
6. Replace the classification layer with a **5-class Softmax layer**.
7. Train the classifier.
8. Fine-tune the final DenseNet layers using a low learning rate.

Loss Function

For five classes:

Categorical Cross-Entropy

Evaluation

Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Justification:

DenseNet is effective because its dense connections promote **feature reuse**, which is especially useful when the available plant-disease training data is limited.

3. A university wants to predict a student's future academic performance using their **weekly attendance, assignment marks, internal assessment marks, and previous examination scores** collected over several weeks. **Design an LSTM-based model** that can process this sequential information and predict the student's performance category.

Answer:

Problem:

The university wants to predict future student performance using weekly attendance, assignment marks, internal assessment marks and previous examination scores collected over several weeks.

Input Data

For every week, the input contains:

[Attendance, Assignment Marks, Internal Marks, Previous Exam Score]

Example:

Week 1 → Week 2 → Week 3 → ... → Week N

Model Architecture

Weekly Data → LSTM → Dropout → Dense → Softmax → Performance Category

For example, the output categories can be:

- Poor
- Average
- Good
- Excellent

Working

The LSTM processes the weekly sequence one time step at a time.

The LSTM maintains:

- **Cell state**
- **Hidden state**

These allow it to remember important information from previous weeks.

For example, consistently decreasing attendance combined with decreasing assignment marks can help the model predict poor future performance.

Training Steps

1. Collect weekly student data.
2. Handle missing values.
3. Normalize numerical features.
4. Convert records into sequences.
5. Divide data into training and testing sets.
6. Train an LSTM model.
7. Use Softmax for multi-class prediction.
8. Optimize using Adam and categorical cross-entropy.

Performance Evaluation

Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Justification:

LSTM is suitable because student information is **sequential**. It can learn relationships between current performance and information from previous weeks.

4. An organization wants to classify customer reviews as **positive, negative, or neutral**. Since the meaning of a word may depend on both the words appearing before and after it, **design a Bidirectional RNN model** for this application. Identify the major processing stages from input text to final sentiment prediction.

Answer:

Problem:

The organization wants to classify customer reviews as **Positive, Negative or Neutral**. The meaning of a word may depend on both preceding and following words.

Major Processing Stages

Review Text

↓

Text Cleaning

↓

Tokenization

↓

Word Embedding

↓

Bidirectional RNN

↓

Dense Layer

↓

Softmax

↓

Positive / Negative / Neutral

Model Architecture

Example:

- Input layer
- Embedding layer
- Bidirectional RNN/GRU/LSTM
- Dropout
- Dense layer
- Softmax output with **3 neurons**

Why Bidirectional RNN?

A normal RNN processes:

Past → Present

A Bidirectional RNN processes:

Past → Present

and

Future → Present

Therefore, the model can understand the context from both sides.

For example:

"The movie was not good at all."

The word "**good**" must be interpreted using the surrounding words, particularly "not".

Training

- Convert text into tokens.
- Convert tokens into embeddings.
- Feed sequences to the BiRNN.
- Use categorical cross-entropy.
- Use Adam optimizer.
- Train using labeled reviews.

Evaluation

Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Justification:

Bidirectional processing provides information from both past and future context, making it suitable for sentiment analysis where word meaning depends strongly on surrounding words.

5. A software company wants to develop a system that translates **English sentences into Hindi**. The length of the input and output sentences may differ, and the system needs to understand the complete input sentence before generating the translated sentence. **Design an Encoder–Decoder model** and describe the role of the encoder, decoder, hidden representation, and output layer in the translation process.

Answer:

Problem:

The company wants to translate English sentences into Hindi. Input and output sentence lengths may differ, and the system must understand the complete input before generating the translation.

Model Architecture

English Sentence

↓

Tokenization + Embedding

↓

Encoder

↓

Hidden Representation / Context

↓

Decoder

↓

Softmax

↓

Hindi Sentence

1. Encoder

The encoder receives the English sentence word by word.

Example:

I → am → going → home

The encoder processes the complete sequence and produces a **hidden representation** containing information about the input sentence.

2. Hidden Representation

The hidden representation acts as a summary/context of the English sentence.

It transfers the information learned by the encoder to the decoder.

3. Decoder

The decoder generates the Hindi sentence **one word at a time**.

Example:

मैं → घर → जा → रहा → हूँ

The previously generated word is used to help predict the next word.

4. Output Layer

The output layer uses **Softmax** to calculate probabilities for possible Hindi words.

The word with the highest probability is selected at each time step.

Training

1. Tokenize English and Hindi sentences.
2. Convert words into numerical representations.
3. Pad sequences.
4. Train the encoder-decoder model.
5. Use teacher forcing during training.
6. Use categorical cross-entropy loss.
7. Optimize using Adam.
8. Evaluate using translation metrics such as BLEU.

Why Encoder–Decoder?

The encoder-decoder architecture is suitable because:

- Input and output lengths can be different.
- The encoder captures the complete input sequence.
- The decoder generates the translated sequence sequentially.
- It can be extended with **attention mechanisms** for better long-sentence translation.

Final Architecture

Encoder:

English → Embedding → LSTM/GRU → Hidden State

Decoder:

Hidden State + Previous Hindi Word → LSTM/GRU → Dense → Softmax → Next Hindi Word

Justification:

The Encoder–Decoder model is designed for sequence-to-sequence tasks such as machine translation, where the input and output sequences can have different lengths.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Requirement Analysis	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Model Architecture & Design	30%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Application of Deep Learning Techniques	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Model Development & Implementation Strategy	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Performance Analysis & Justification	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis